

WebRTC implementation In chat app

What is WebRTC?

WebRTC (Web Real-Time Communication) is a browser technology that enables direct peer-to-peer audio, video, and data communication between browsers without going through a server.

2. Key WebRTC Concepts

A. RTCPeerConnection

The main object that manages the connection between two browsers

```
const peerConnection = new RTCPeerConnection(configuration);
```

.

B. MediaStream (getUserMedia)

Accesses the user's camera and microphone.

```
const stream = await navigator.mediaDevices.getUserMedia({  
  audio: true,  
  video: true  
});
```

C. SDP (Session Description Protocol)

A text format that describes media capabilities (codecs, resolution, etc.).

-OFFER

-accept

D. ICE Candidates

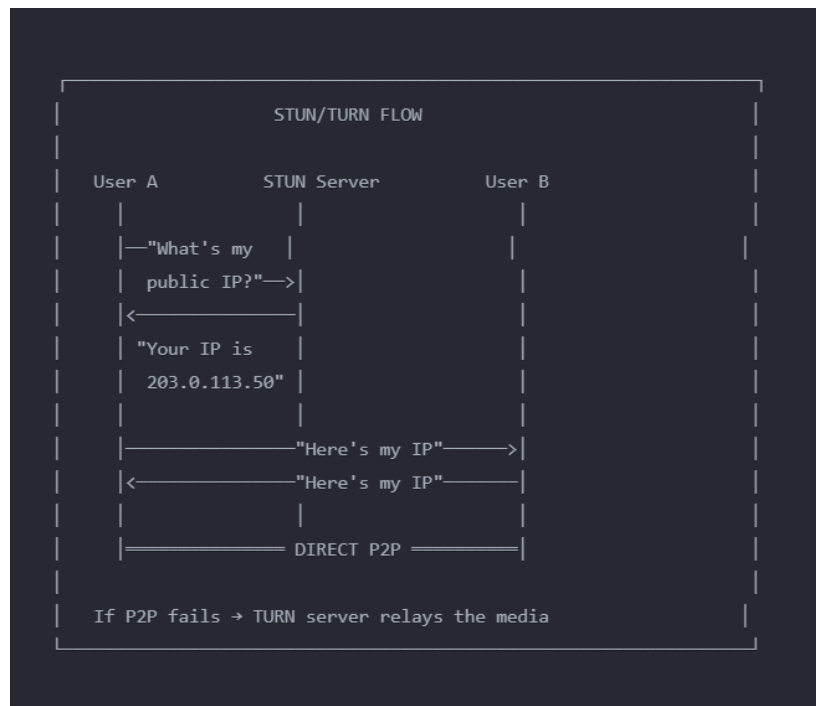
Network connection information (IP addresses, ports) that helps browsers find the best path to connect.

Copy

candidate:1234567890 1 UDP 2122260223 192.168.1.100 54321 typ host
candidate:1234567891 1 UDP 1686052607 203.0.113.50 12345 typ srflx

E. STUN/TURN Servers

Server Type	Purpose	Example
STUN	Discovers your public IP address	<code>stun:stun.l.google.com:19302</code>
TURN	Relays media when direct P2P fails (firewalls/NAT)	<code>turn:your-turn-server.com:3478</code>



WebRTC Signaling Flow



My chat app

Users

Connected

Search

tom tom@gmail.com



Chat with tom

You
tom

2020-04-29 22:23:47

You
what

2020-04-29 22:23:51

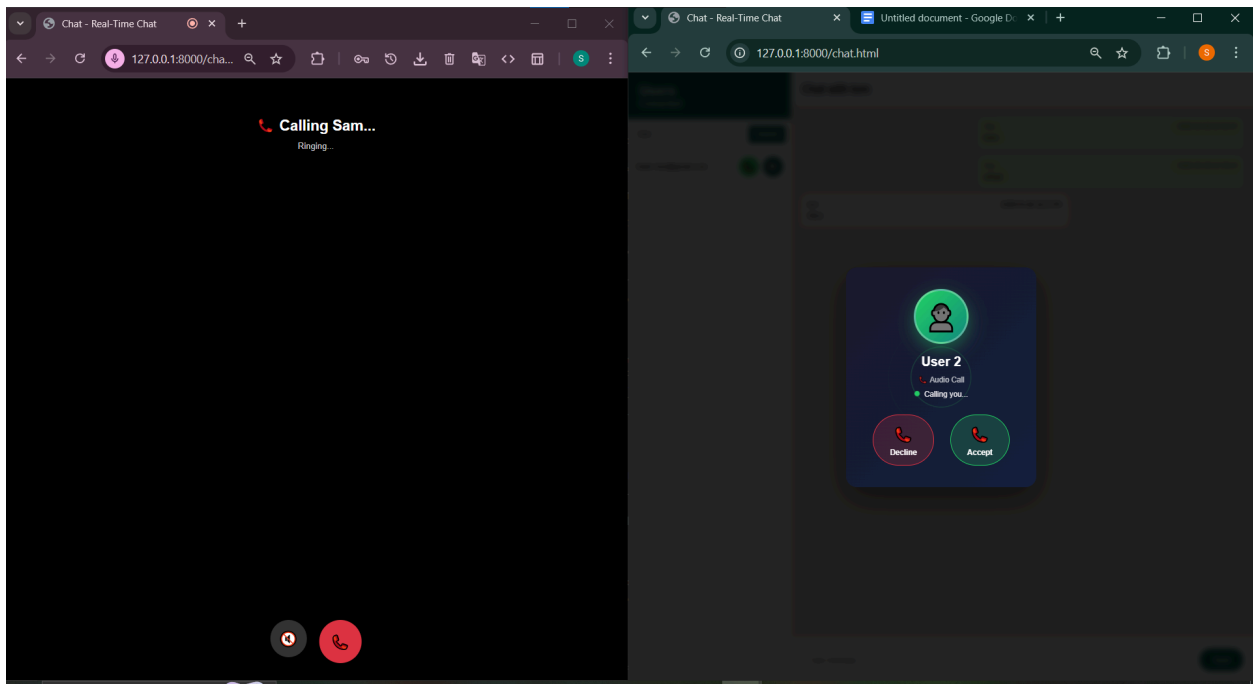
tom
nice

2020-04-29 22:23:54

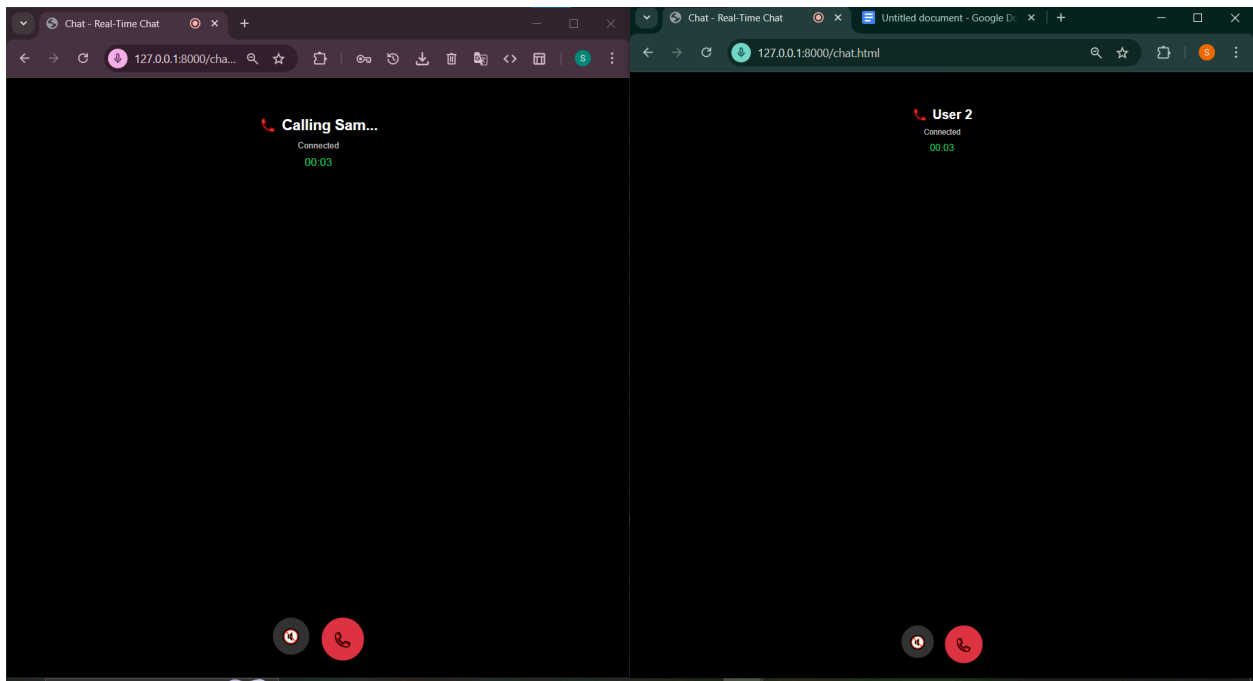
Type message...

Send

When user Tom click on call button



After user 2 accept call:



Similar for video call video is shown on both side when requested call is accepted by other

Feature:

. User Authentication & Authorization

- JWT-based authentication system
- Secure password hashing (bcrypt)
- Token-based session management
- Protected routes requiring authentication
- Automatic token validation on WebSocket connections
- User login/logout functionality
- Token stored in localStorage for persistence

Real-Time Messaging

- WebSocket-based instant messaging
- Private one-on-one chat between users
- Message delivery confirmation
- Real-time message synchronization
- Message timestamp tracking
- Support for text messages
- Sent/received message visual distinction
- Auto-scroll to latest message
- Enter key to send messages

User Management

- User search functionality
- Display all registered users
- Online/offline user status detection
- User join/leave notifications
- User profile display (username, email)
- Click-to-select user for chat
- Active user list in sidebar

Chat History

- Per-user chat history storage
- Persistent message storage in database
- Load previous conversations on user selection
- Message history with timestamps
- Sender/receiver identification
- SQLAlchemy ORM for database operations
- PostgreSQL/SQLite database support

WebRTC Audio Calling

- One-to-one audio calls
- In-page call overlay (no redirects)

- Call initiation with ringing state
- Incoming call notification modal
- Accept/Reject call functionality
- Real-time audio streaming
- Mute/unmute audio during call
- Call duration timer
- Call end cleanup

WebRTC Video Calling

- One-to-one video calls
- Picture-in-picture local video preview
- Full-screen remote video
- Camera on/off toggle during call
- Video call overlay matching WhatsApp style
- Automatic camera/mic cleanup on call end
- Call state management (ringing, connected, ended)

Call Management

- Call history logging (caller, receiver, duration, status)
- Call status tracking (initiated, answered, missed, rejected, ended)
- Call type tracking (audio/video)
- Database storage for call logs
- Call notification to receiver before acceptance
- Both users see video only after call is accepted
- Proper media stream cleanup

User Interface

- WhatsApp-inspired design
- Responsive chat layout
- Sidebar with user list
- Search box for finding users
- Call buttons (audio/video) per user
- Incoming call modal with accept/reject
- Active call overlay with controls
- Connection status indicator
- Clean, modern UI with CSS animations
- Ringing animation for incoming calls

WebSocket Management

- ConnectionManager class for WebSocket handling
- User connection tracking
- Automatic disconnect handling

- Broadcast messaging to all users
- Targeted messaging to specific users
- WebRTC signaling through WebSocket
- ICE candidate exchange
- SDP offer/answer exchange
- Connection state monitoring

WebRTC Features

- STUN server configuration (Google STUN)
- RTCPeerConnection management
- MediaStream API (getUserMedia)
- SDP offer/answer exchange
- ICE candidate negotiation
- Direct peer-to-peer media streaming
- Connection state monitoring
- Automatic reconnection handling

Call Controls

- Mute/unmute microphone
- Camera on/off toggle
- End call button
- Call timer display
- Connection status display
- Visual feedback for controls
- Button state management